. Worksheet No. 1

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EXPLORING QUADRATIC FUNCTIONS OF THE FORM $y = ax^2 + bx + c$

Group:.....

Read carefully and fill in the blanks:

Task 1: Visualise the curve's shape

Step 1: Scan QR Code below to access GeoGebra Classroom.

Step 2: On the left panel, click the "Algebra" tab (the calculator icon).

Step 3: Click the input bar and type: $y = ax^2 + bx + c$ then press "Enter".

Step 4: Move the sliders for b and c to any value.

Step 5: Now, focus on slider a. Slide it and observe the change:

- When $a > 0$, the graph is $\cup$-shaped.
- When $a < 0$, the graph is $\cap$-shaped.

Task 2: Consider the graph $y = 2x^2 + 4x - 6$, follow the following steps to find the essential information for sketching the graph:

1. Shape: $a = 2 > 0$, so the graph is $\cup$-shaped.
2. y-intercept: Set $x = 0$ then $y = -6$. Coordinate: (0,6).
3. x-intercepts: Set $y = 0$ then we obtain $2x^2 + 4x - 6 = 0$.

- Factorize: $2(x-1)(x+3) = 0$.
- $x = 1$ or $x = -3$. Coordinates: (1, 0) and (-3, 0).

4. Turning Point:

- Find axis of symmetry: $x = -\frac{b}{2a} = -\frac{4}{2.2} = -1$.
- Substitute $x = -1$ to find $y = 2.(-1)^2 + 4.(-1) - 6 = -8$.
- Turning point coordinate: $(-1, -8)$.

Task 3: Receive AI feedbacks

1. Scan QR Code to open the AI Chatbot.

2. Take a clear photo of BOTH **Task 1** and **Task 2** and upload them to the chatbot.
3. To ask the AI to check your work, please type the prompt: *"Please check my observations in Task 1 and my calculations in Task 2 for* $y = 2x^2 + 4x - 6$. *Give me hints if there are any errors."*
4. Write down the AI's feedbacks here:

"Your observations in Task 1 are correct. You correctly identified that when $a > 0$, the graph is $\cup$-shaped, and when $a < 0$, the graph is $\cap$-shaped.

In Task 2, your calculations are accurate. You correctly found the y-intercept, the x-intercepts, and the turning point of the graph. Your method is clear and logically presented.

Well done! Keep up the good work."

5. Correct your answer based on the AI's feedbacks.

Task 4: Find the turning point by completing the square

You can find the coordinates of the turning point of a parabola algebraically by changing the equation from $y = ax^2 + bx + c$ *into the form* $y = a(x+p)^2 + q$. *In this form, the turning point is* $(-p, q)$.

For example, let us consider the equation: $y = 2x^2 + 8x - 8$.

Method 1: By formula

Step 1: Identify the coefficients from the standard form $y = ax^2 + bx + c$:

$a = 2$, $b = 8$, $c = -8$.

Step 2: Find the x-coordinate using the formula $x = -\frac{b}{2a}$: $x = -\frac{8}{2.2} = -2$.

Step 3: Find the y-coordinate by substituting the value of x into the original equation: $y = 2.(-2)^2 + 8.(-2) - 8 = -16$.

Step 4: Based on calculation, the turning point is: (-2,-16).

Method 2: By completing the square

Consider the equation $y = 2x^2 + 8x - 8$

This can be rewritten as $y = 2(x+2)^2 - 16$ by completing the square.

Squaring any value results in an answer that is either positive or 0. This means that for any value of x, the smallest value of $(x+2)^2$ is 0.

This means that the minimum value of $y = 2(x+2)^2 - 16$ is -16 and that this occurs when $x = -2$.

The turning point of the graph $y = 2(x+2)^2 - 16$ has the coordinates: $(-2, -16)$.